

FUZZY LOGIC FOR AUTONOMOUS SPACECRAFT ATTITUDE CONTROL FAILURE

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Summary: Spacecraft failures occur in many ways. In the worst case they can even provoke the total loss of the mission. Some failures can seriously affect the attitude control system performance. In the majority of cases, attitude control failure recovery is a laborious manual process. The paper presents the advantages of fuzzy control methodology applied to autonomous system failure recovery. The fuzzy controller is especially useful when the plant is difficult to model mathematically, e.g., after a failure has occurred. The paper discusses the sequence of events in the implementation of the fuzzy controller: acquisition and capture of the knowledge of an experienced human spacecraft controller, risk analysis, elaboration of the rules data base, construction of the inference engine, and the final implementation in a microprocessor.

1. Introduction

A spacecraft attitude control system is the component part of a spacecraft in charge of measuring its orientation in space. It calculates the actual state of the vehicle and computes the desired control forces to maintain a determined orientation or acquire a new one. It contains several blocks (figure 1): the navigation block calculates the actual state of the vehicle; the guidance part calculates the future state of the spacecraft to achieve the desired trajectory and the control part calculates the desired control torques to achieve this trajectory.

The objectives are to maintain the vehicle within a prescribed orbit and attitude having fuel consumption and maneuver time minimized. The main control requirements of a spacecraft are formulated as a deviation from conditions of regular orientation.

The motion equations of the spacecraft are nonlinear [6]. The performance of sensors and actuators is not totally perfect, and the size of the spacecraft produces elastic modes neglected in the mathematical model of the plant. In most occasions, low and high frequencies appear with very low damping.

Bode diagrams and phase plots are insuffi-

cient to forecast totally the plant behavior in all circumstances, and approximation in the discretisation process must be done carefully. In case of a failure, the situation deteriorates more, and the knowledge of the plant is very limited. In this case, there is a significant degree of fuzziness.

Spacecraft failures occur in many different ways [5]. Often, the anomalies occur during the early launch stages. Launcher failures, appendage deployment anomalies, propulsion system errors, etc., are the most common causes of problems in the life time of a spacecraft. If the failing system has been designed with a redundant unit, the spacecraft control system can switch from the failing unit to the healthy one to be

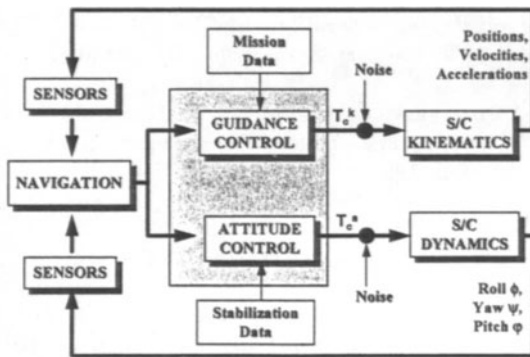


Figure 1: Spacecraft Control System

able to continue normal operations. If the failing system is not or could not be designed with a backup unit (apogee motor, etc.), a failure can cause the loss of the mission.

Potentially all failures can affect the functioning of the attitude control system, causing the spacecraft to deviate from its nominal orientation. Moreover, some failures can be overcome changing the performance and modus operandi of the attitude control system.

In 1989, ESA officials managed to resort the spacecraft HIPPARCOS to a revised mission to be able to achieve its objectives. The apogee engine suffered a failure which prevented the spacecraft to achieve the final geostationary orbit. The satellite had to orbit in an inclined elliptic trajectory while trying to catalogue more than 300.000 stars. Changing the parameters of the attitude and orbit control system allowed HIPPARCOS to measure a significant part of the foreseen mission. In 1991, a series of electrical and mechanical problems caused the geostationary communications satellite OLYMPUS to go out of control for two months, until engineers managed to put the spacecraft back to a suitable position and orientation. In all of these cases, the reposition and reorientation of the spacecraft is a delicate task, performed on ground, and supervised by large teams. Inevitably, this leads to the consumption of a large amount of resources: time and money.

2. Fuzzy Spacecraft Control

Fuzzy logic has shown to be specially suitable in occasions when the plant is not static but changes with time (or differs slightly among very similar systems), or when the characteristics of the plant are not totally known or understood at the time when the controller was designed, or when the control actions and goals were not precisely defined. Fuzzy logic has been proven to be adequate to solve control problems not in the *best* way, but just in a *suitable* way within the required limits, and giving satisfactorily performance.

Two strategies that can be used in fuzzy spacecraft control: *direct* control and *supervisory* control. Both types of control are called *expert*, because they incorporate the knowledge of an expert, embedded in the control laws. Direct control can be used when the

spacecraft control system is managed by a centralized control law. Supervisory control (figure 2) is adequate when the control system is hierarchically constructed, having several control laws for different sensors and actuators. Nowadays, the hierarchical approach is dominant in the modern type of spacecraft design [3],[8].

The configuration of most spacecraft exhibits the following characteristics: the spacecraft is not a rigid body but an object with multiple moving appendages. The final mass is not known with total precision, until the complete spacecraft is finished and filled up with fuel (e.g. time close to the launch); so the control system must be designed with certain tolerances [9]. A satellite thruster system can never be perfectly aligned; at the beginning of the life of the satellite every maneuver has to be carefully calibrated. Once in its final position, the movement of the solar arrays provoke structural flexures to the spacecraft dynamics [10], [11]; as a consequence, structural resonances can occur disturbing the attitude. In most occasions, when thrusters are fired (reorbit or station keeping), the satellite experiences parasitic torques along all axis, different from the one containing the fired device. As time passes, the fuel consumption varies the total mass of the satellite, and therefore the centre of gravity changes. The matrix of inertia is not diagonal: there are cross products of inertia [12].

All these drawbacks with a high degree of uncertainty, suggest the use of fuzzy logic as an alternative to the crisp control system design and implementation. A big beneficial consequence of the application of

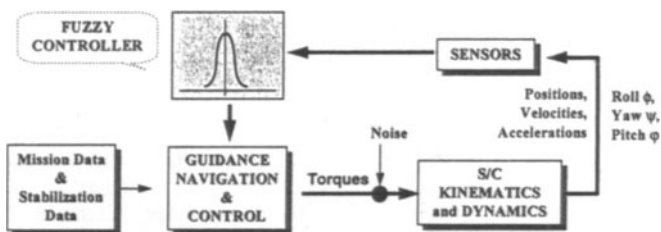


Figure 2: Supervisory Control